

ECE4010J Probabilistic Methods in Engineering (Summer 2026)

Assignment 9

Date Due: See canvas

This assignment has a total of **(115 points)**.

Note: Unless specified otherwise, you must show the details of your work via logical reasoning for each exercise. Simply writing a final result (whether correct or not) will receive **0 point**.

Exercise 9.1 (10 pts) A study is conducted to see if there is an association between age and willingness to use computerized banking systems. The data shown below are obtained in a survey of 500 randomly selected customers of a bank that has been offering computerized banking for over a year. Is there evidence of an association between these two variables? Explain, based on the p -value of your test and inspection of the table.

Age	Use computerized banking	
	Yes	No
Under 40	150	75
40 or over	150	125
		500

Exercise 9.2 (10 pts) A new method for etching semiconductors is being studied. The quality of the etch is to be compared to that obtained using two older techniques. The results of the study are given below. State the null hypothesis of homogeneity mathematically. Test this hypothesis at the $\alpha = 0.05$ level. Interpret your result in a practical sense.

Method	Quality			
	Excellent	Good	Fair	Poor
High pressure (old)	113	34	21	32
Reactive ion (old)	117	31	25	27
Magnetron (new)	130	40	20	10
				600

Exercise 9.3 (25 pts) The relationship between energy consumption and household income was studied, yielding the following data on household income X (in units of \$1000/year) and energy consumption Y (in units of 10^8 Btu/year).

Energy consumption (y)	Household income (x)
1.8	20.0
3.0	30.5
4.8	40.0
5.0	55.1
6.5	60.3
7.0	74.9
9.0	88.4
9.1	95.2

- (i) (5 pts) Estimate the linear regression equation $\mu_{Y|x} = \beta_0 + \beta_1 x$.
- (ii) (5 pts) If $x = 50$ (household income of \$50000), estimate the average energy consumed for households of this income. What would your estimate be for a single household?
- (iii) (5 pts) How much would you expect the change in consumption to be if any household income increases \$2000/year (2 units of \$1000)?
- (iv) (5 pts) How much would you expect consumption to change if any household income decreases \$2000/year?
- (v) (5 pts) Write the normal equations for these data.

Exercise 9.4 (30 pts) In production flow-shop problems, performance is often evaluated by minimum make-span, the total elapsed time from starting the first job on the first machine until the last job is completed on the last

machine. For a particular flow-shop the make-span was evaluated with respect to the number of jobs to be done. Let the independent variable X denote the number of jobs and the dependent variable Y denote the make-span (in standardized units):

Number of jobs (x)	4	5	6	7	8	9	10	11	12	13	14	15
Make-span (y)	3.75	4.90	4.88	7.2	7.3	9.1	9.0	11.9	11.5	14.1	13.9	17.5

- (5 pts) Estimate the linear regression equation $\mu_{Y|x} = \beta_0 + \beta_1 x$.
- (5 pts) Plot the estimated regression equation.
- (5 pts) Test for a significant linear regression at the $\alpha = 0.05$ level of significance.
- (5 pts) At $x = \bar{x}$, compute a 95% confidence interval for $\mu_{Y|x}$, and verbally explain the answer.
- (5 pts) At $x = 12$, compute a 95% confidence interval for $\mu_{Y|x}$, and verbally explain the answer.
- (5 pts) How do you explain the different widths of the intervals in parts (iv) and (v)?

Exercise 9.5 (20 pts) A study reported in the *Journal of Coatings Technology*, vol. 55, 1983, considered the ability to predict cracking of latex paints on exposed wood surfaces based on accelerated cracking tests. The following are representative data on accelerated crack rating (x) and exposure crack rating (y):

Accelerated crack rating (x)	Exposure crack rating (y)
2.0	1.9
2.0	2.3
3.0	2.7
3.0	3.9
4.0	3.0
4.0	4.2
5.0	3.1
5.0	4.8
6.0	4.8
6.0	6.7
7.0	5.1
7.0	6.4

- (5 pts) Estimate the line of regression for predicting exposure crack rating from accelerated crack rating.
- (5 pts) Estimate the average exposure crack rating if the accelerated crack rating is 4.5.
- (5 pts) Test for lack of linear fit at the 0.05 level of significance.
- (5 pts) Can we conclude that a linear regression equation adequately fits the data?

Exercise 9.6 (20 pts) Pesticides used in food production can be found in food consumed by humans. A study focusing on chickens exposed to malaoxon was conducted. The chickens were also exposed to a liver enzyme inducer to determine whether liver detoxification of the pesticide is affected. The following data were reported as a percentage of normal pesticide detoxification (y) and percentage of normal liver enzyme levels (x):

Enzyme level (x)	Detoxification level (y)
95	108
110	126
118	102
124	121
145	118
140	155
185	158
190	178
205	159
222	184

- (5 pts) Estimate ρ , the correlation between X and Y .

- (ii) (5 pts) Test the null hypothesis that X and Y are uncorrelated at the 0.10 level. That is, test $H_0 : \rho = 0$. Discuss your conclusion.
- (iii) (5 pts) Find a 90% confidence interval on ρ .
- (iv) (5 pts) Test $H_0 : \rho = 0.8$ at the $\alpha = 0.05$ level of significance.